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DNS Log Analysis Lab Report

Using Splunk Basics – DNS Log Analysis Lab Report

Structured Cybersecurity Report

OBJECTIVE

The main objective of this lab is to analyze DNS log data using Splunk and understand how domain-related network activity can be monitored.

In this lab, DNS logs are ingested into Splunk and analyzed using SPL queries to identify patterns in network traffic. The analysis focuses on understanding which domains are being queried, which systems are generating the most DNS requests, and what types of DNS queries are being used.

This helps in gaining basic knowledge of how DNS activity can be used to detect unusual behavior or potential security issues in a network.

TOOLS AND ENVIRONMENT

This lab was performed using **Splunk**, which acts as the primary tool for searching, monitoring, and analyzing log data. It allows efficient handling of large volumes of network logs and helps in identifying patterns in DNS activity.

The dataset used in this lab consists of **DNS logs generated by Zeek**, a network monitoring tool that records detailed information about domain name resolution requests made by clients within the network.

These logs are stored in **JSON format**, which is a structured data format. Because of this, Splunk is able to automatically extract important fields without requiring manual configuration.

Key Details of the Environment:

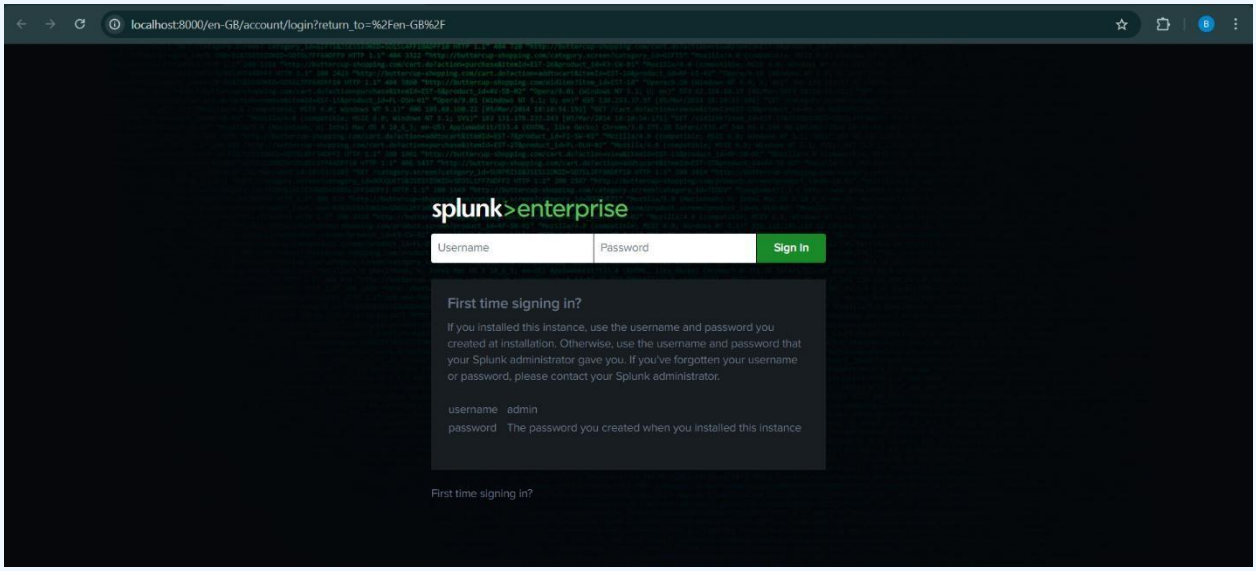
- **Tool Used:** Splunk (for log analysis and visualization)
- **Log Source:** Zeek DNS logs (captures DNS query activity)
- **Log Format:** JSON (structured format for easy parsing)

LAB SETUP

Before starting the analysis, the DNS log file must be uploaded to Splunk.

Step 1 – Open Splunk Web

1. Open the **Splunk Web Interface** in your browser.
2. Log in using your credentials.

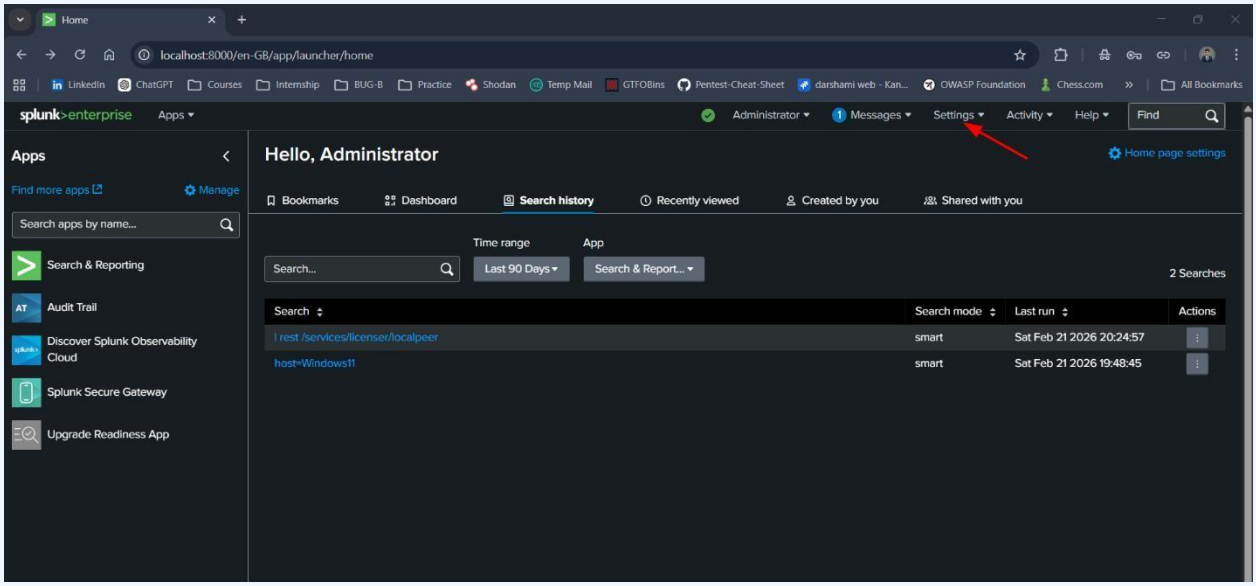


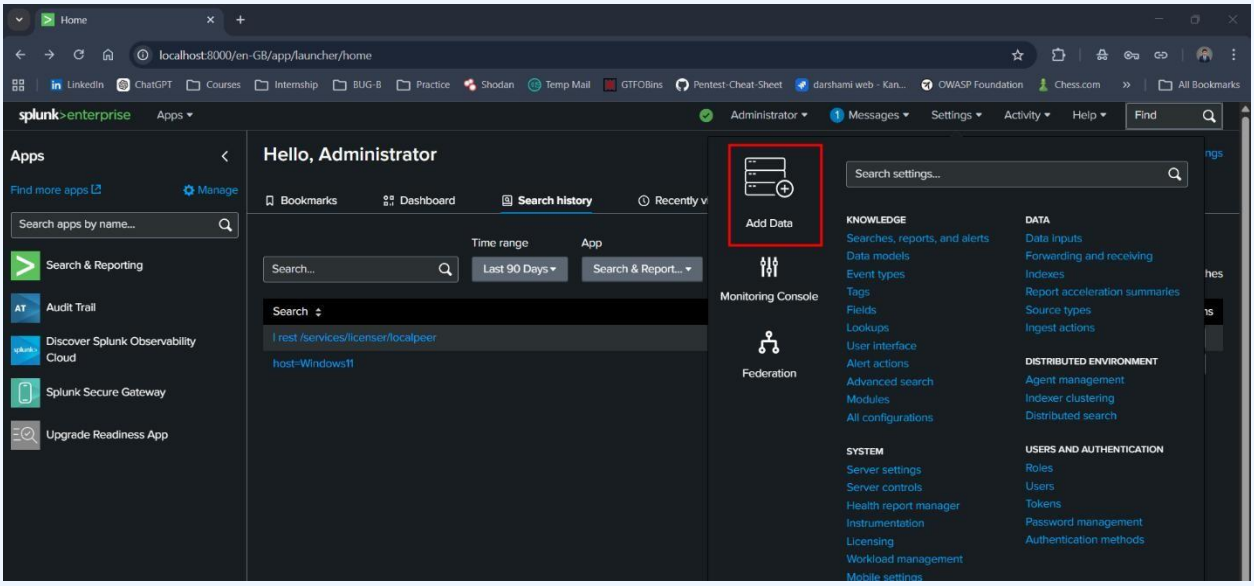
Step 2 – Add Data

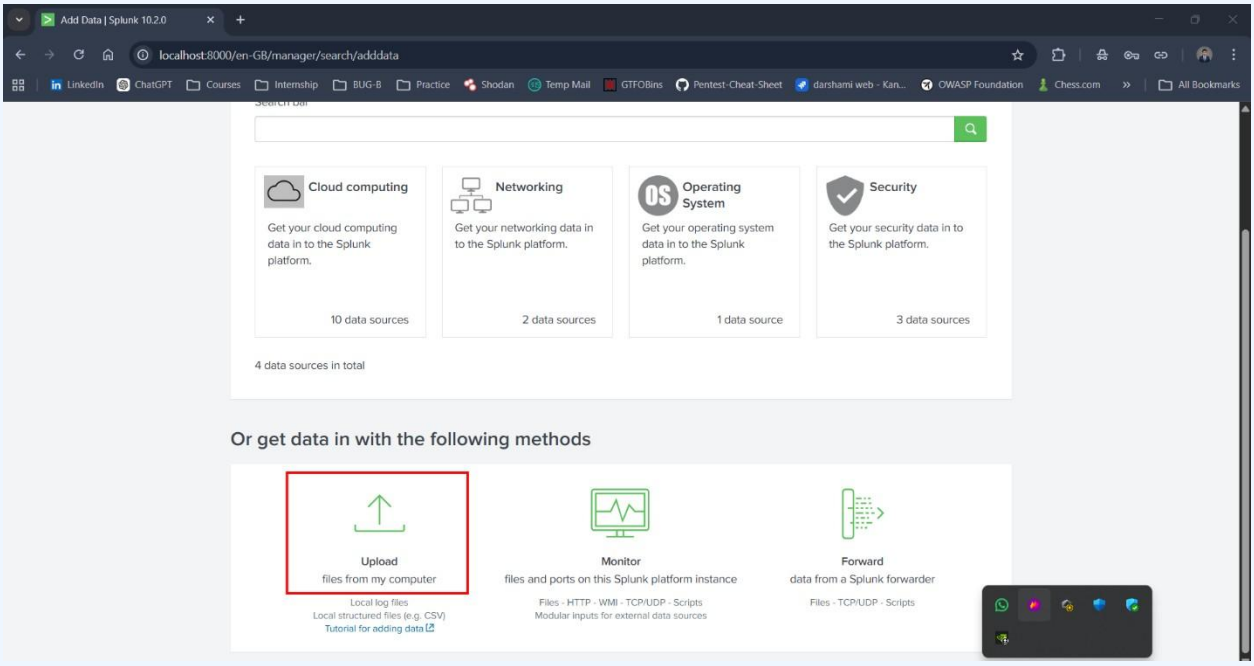
1. Navigate to:

Settings → Add Data

2. Select **Upload.**





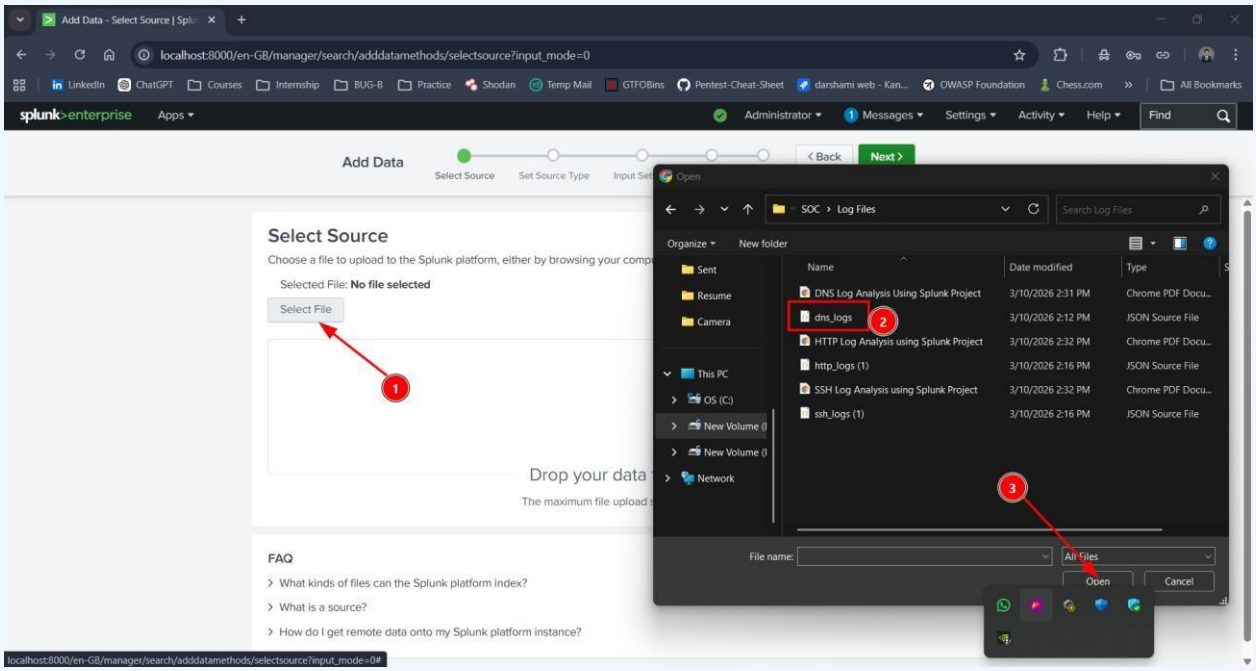


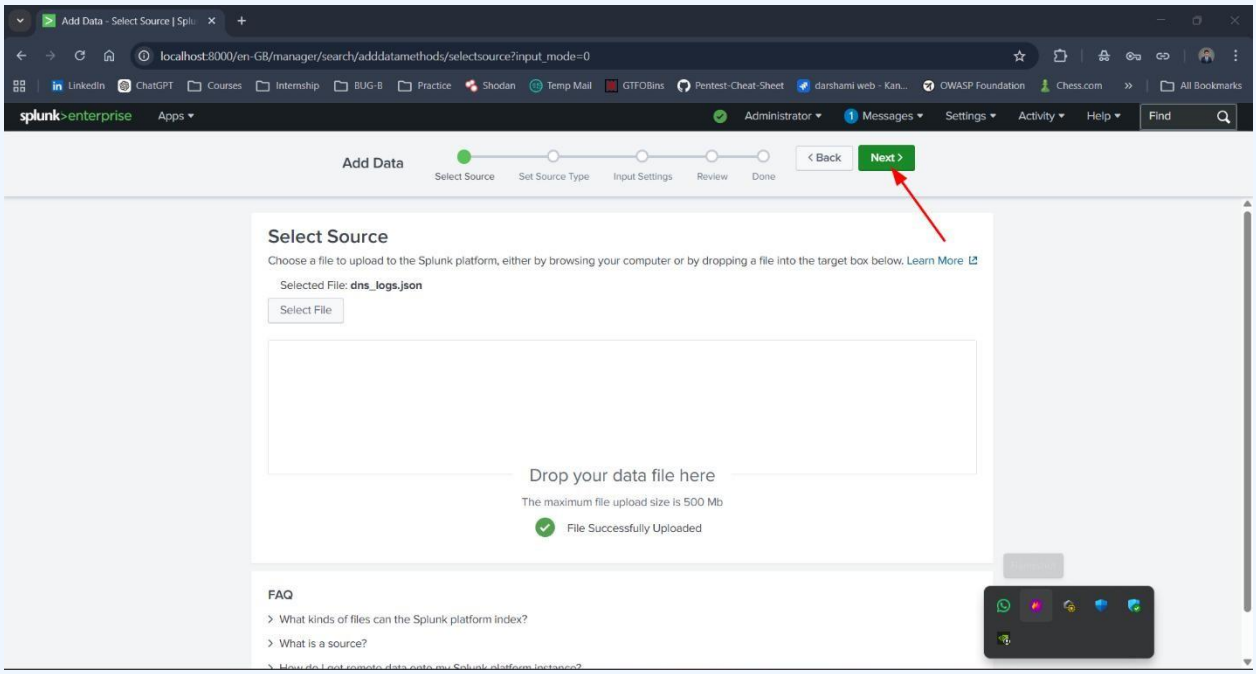
Step 3 – Upload DNS Log File

1. Click **Select File.**
2. Upload the file:

```
dns.log
```

This file contains **DNS query records generated by Zeek network monitoring tool**.





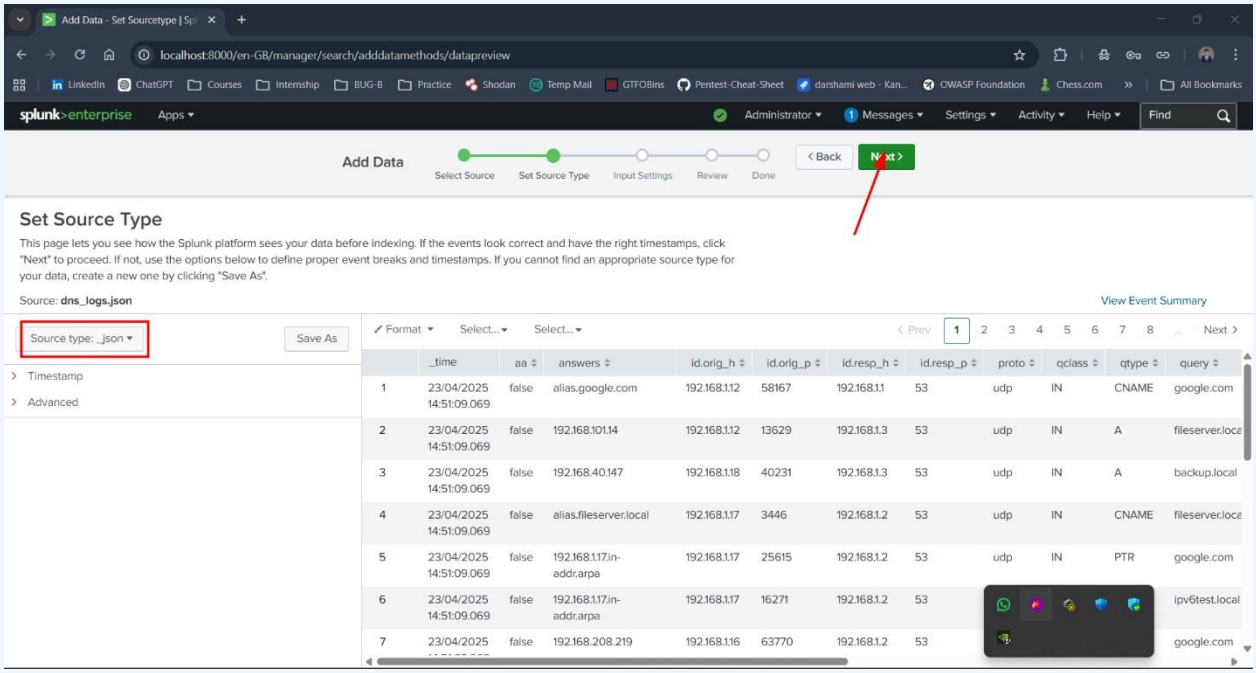
Step 4 – Configure Source Type

The source type was set to **json** to ensure proper parsing of structured log data.

_json

Reason:

- DNS logs are in JSON format
- Splunk automatically extracts fields such as:
    - query
    - id.orig_h
    - qtype
    - answers



Step 5 – Configure Index

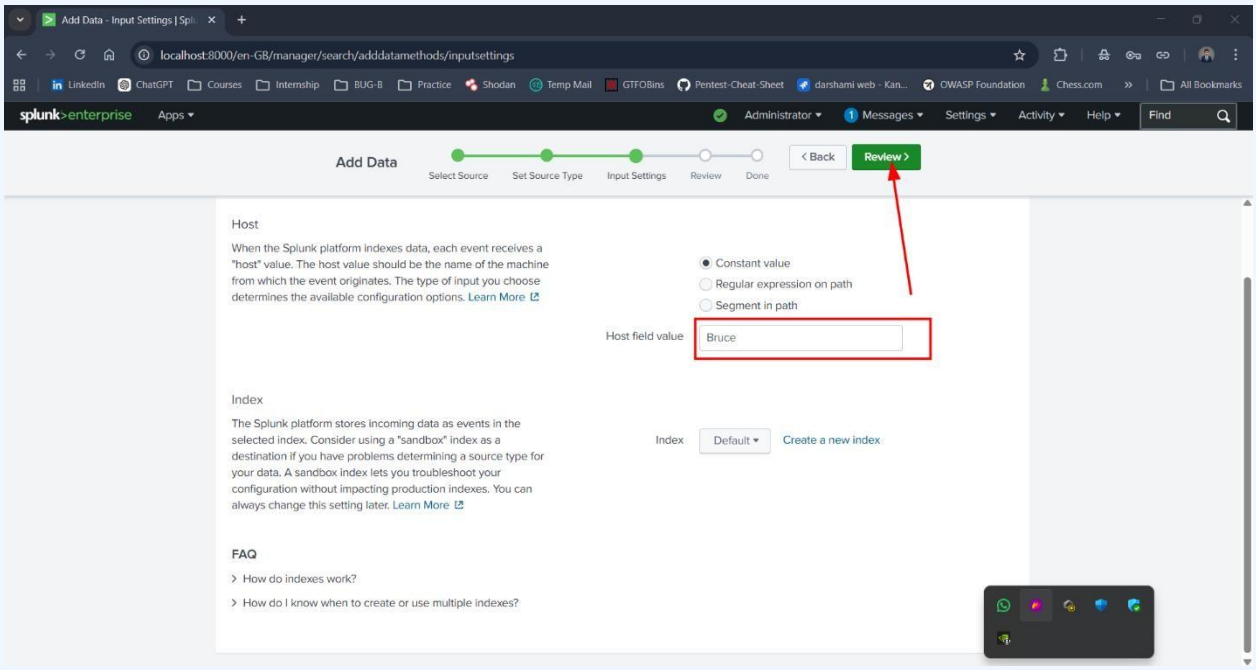
Choose or create an index.

Example:

```
dns_lab
```

Reason:

- Keeps DNS data separate from other logs
- Makes searching and analysis easier



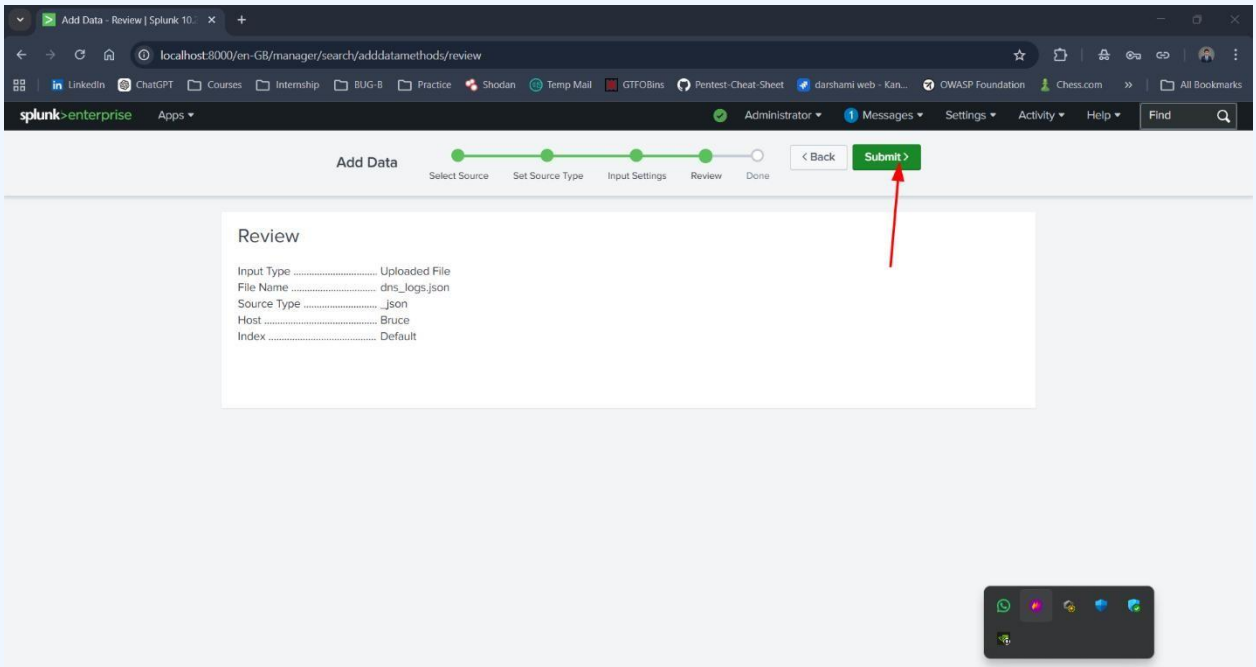
Step 6 – Review and Submit

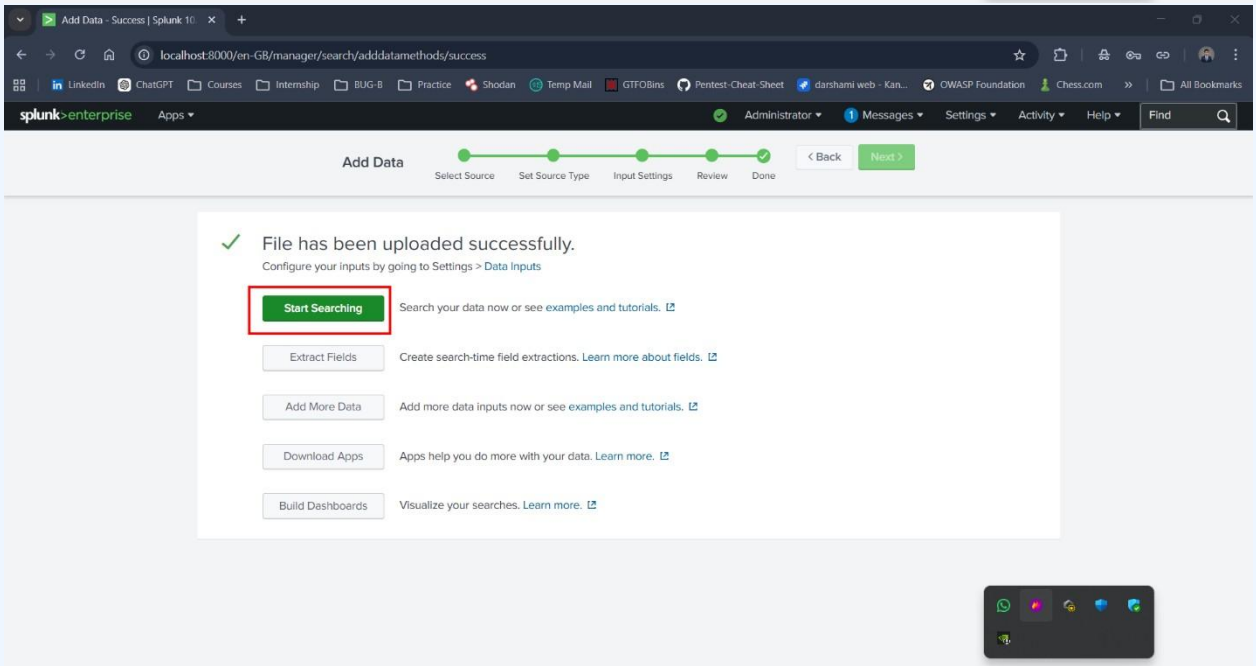
1. Review configuration.
2. Click **Submit**.
3. Splunk will index the data.

You can confirm ingestion by running:

```
index=dns_lab
```

If events appear, the upload was successful.





UNDERSTANDING DNS LOGS

DNS logs record requests made by clients to resolve domain names into IP addresses.

Each entry provides details about which domain was requested, from which IP, and what type of query was made.

Important Fields

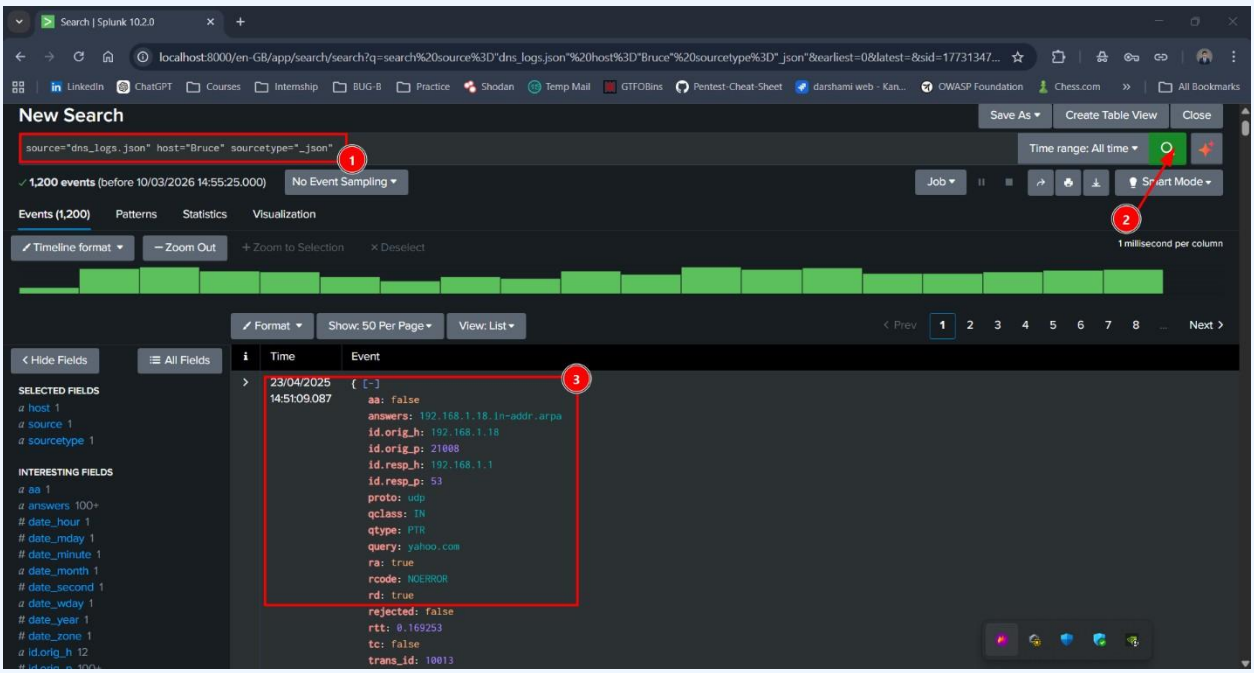
- **id.orig_h** – Source IP making the request
- **query** – Domain name requested
- **qtype** – Type of DNS query
- **answers** – IP address returned

Example

- Source IP: 192.168.1.10
- Domain: google.com
- Query Type: A
- Response: 142.250.190.78

This shows that a client requested the IP address of a domain.
Example Log:

```
{
 “id.orig_h” : “192.168.1.10” ,
 “query” : “google.com” ,
 “qtype” : “A” ,
 “answer” : “142.250.190.78”
}
```



LAB TASKS

Task 1: Identify Frequently Queried Domains

SPL Query:

```
index=dns_lab sourcetype="json"
| stats count by query
| sort -count
```

Explanation

This query is used to analyze DNS logs and identify the most frequently requested domain names in the network.

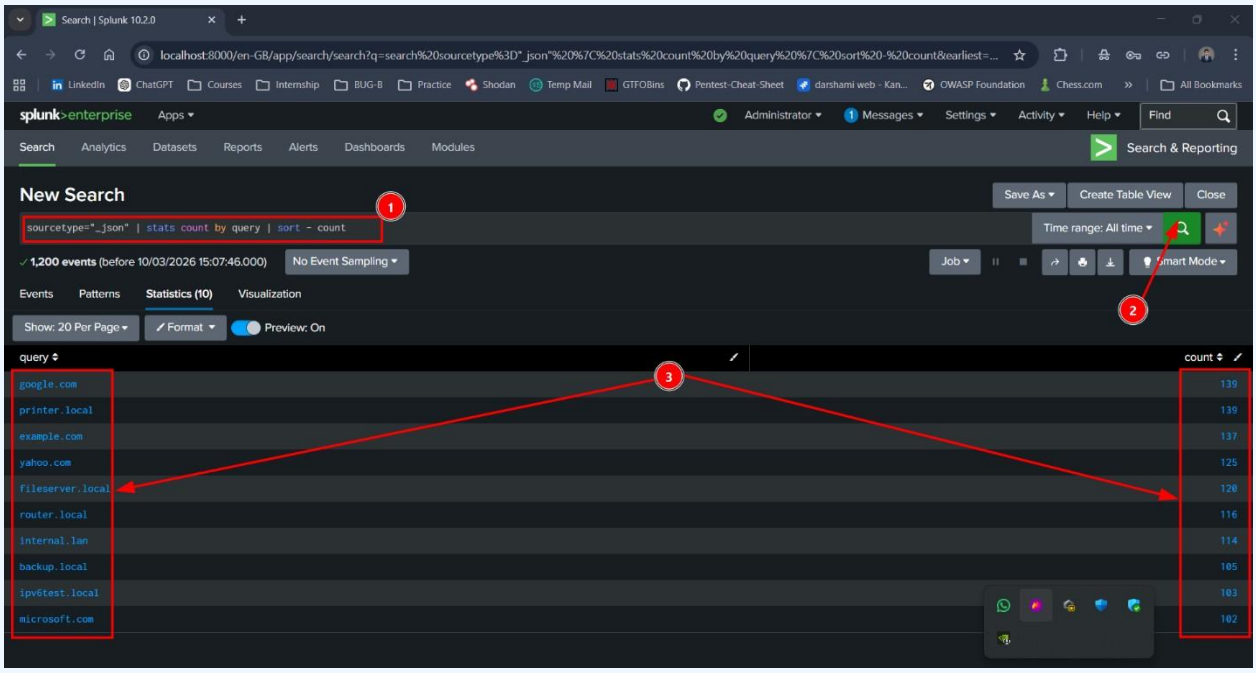
- index=dns_lab
  Searches the DNS logs stored in the specified index.
- sourcetype="json"
  Filters the data to include only JSON formatted log entries.
- stats count by query
  Counts how many times each domain name has been requested.
- sort -count
  Sorts the results in descending order to display the most queried domains at the top.

Analysis Result:

This query helps in understanding domain usage within the network.

From the results, it becomes possible to identify:

- Frequently accessed or popular websites
- Domains that may appear unusual or suspicious
- Patterns in DNS traffic that may require further investigation



Task 2: Find the Most Active User IPs Generating DNS Traffic

SPL Query

```
index=dns_lab sourcetype="json"
| stats count by id.orig_h
| sort -count
```

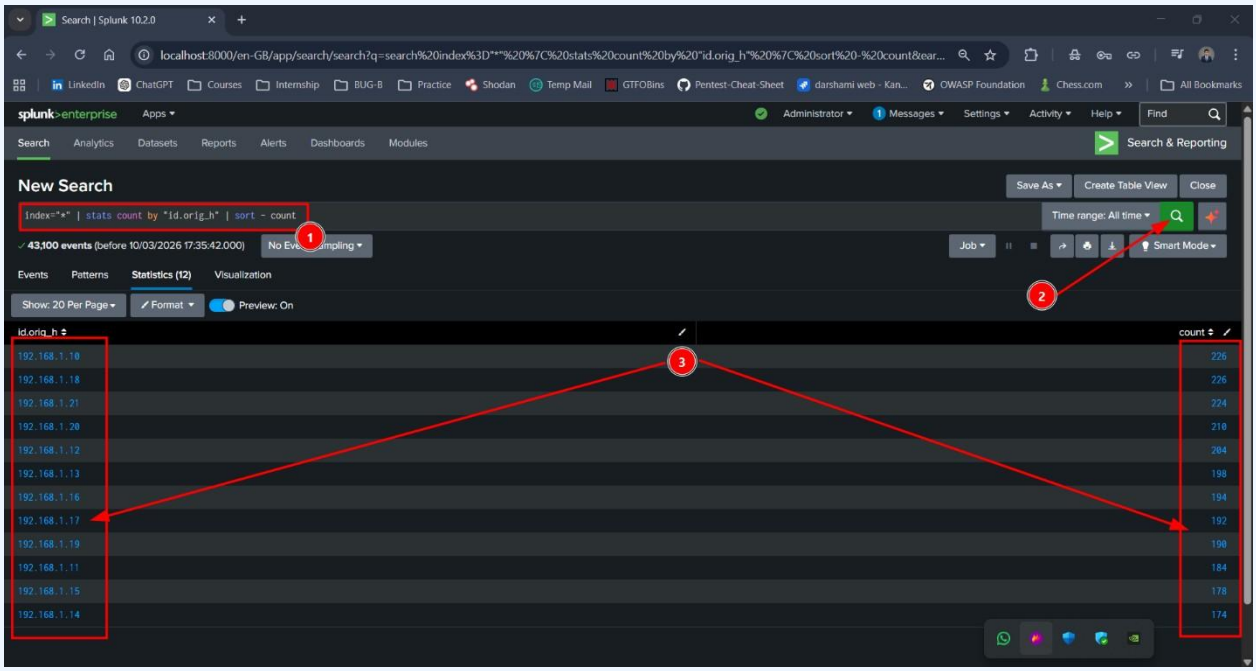
Explanation

This query is used to analyze DNS activity and identify the source IP addresses generating the highest number of requests.

- stats count by id.orig_h
  Counts the total number of DNS queries generated by each source IP address.
- sort -count
  Sorts the results in descending order to display the most active IPs first.

Analysis Result:

- Identifies systems generating high DNS traffic
- Helps detect abnormal behavior from specific hosts
- High activity may indicate automated processes or misconfiguration



Task 3: Breakdown of DNS Query Types

DNS queries can request different types of records.

SPL Query

```
index=dns_lab sourcetype="json"
| stats count by qtype
```

Explanation

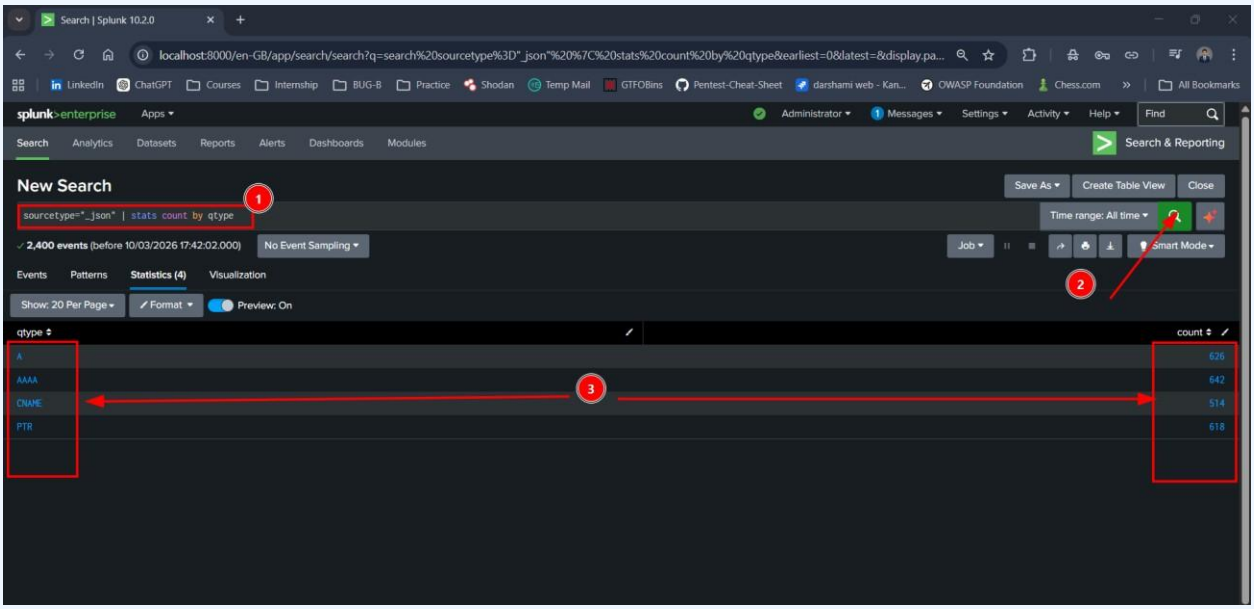
This query groups DNS requests based on their query type.

Common Query Types

- **A** – Resolves domain to IPv4 address
- **AAAA** – Resolves domain to IPv6 address
- **CNAME** – Alias for another domain
- **PTR** – Reverse lookup

Analysis Result

- Shows distribution of query types
- Helps understand how DNS is being used
- Useful for detecting unusual query patterns



SECURITY INSIGHT

DNS log analysis can help in identifying different types of suspicious activities:

- **Malware Communication**
  Systems may connect to unknown domains for command and control

- **DNS Tunneling**
  Data may be transferred using DNS requests

- **DGA Domains**
  Automatically generated domain names used by malware

- **Data Transfer Activity**
  Repeated DNS queries may indicate hidden communication

Indicators to Watch

- Very long or unusual domain names

- High number of requests from a single host

- Queries to unknown or suspicious domains

Splunk helps in quickly identifying such patterns through search and analysis.

CONCLUSION

This type of analysis is essential for **network monitoring, threat detection, and incident response** in cybersecurity environments.

In this lab, DNS logs were successfully uploaded and analyzed using Splunk.

In this lab we successfully:

- Uploaded **Zeek DNS logs into Splunk**
- Learned how Splunk parses **JSON structured data**
- Used **SPL queries** to analyze DNS traffic

The analysis helped in understanding how DNS traffic works and how different queries can be examined using SPL.

During the process, it was possible to identify:

- Frequently queried domain names
- Systems generating high DNS traffic
- Different types of DNS queries

This type of analysis is useful for monitoring network activity and detecting unusual behavior in a system.